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INSPECTION PROCEDURE FOR PRESSURE VESSELS

FWEL CONTRACT No : 1-14-16601/19

CONTRACT NO. : POGC-760-81-156

PROJECT : SOUTH PARS GAS FIELD DEVELOPMENT (PHASES 9&10)

COMPANY : NATIONAL IRANIAN OIL COMPANY

SITE : ASSALUYEH / SOUTH PARS GAS FIELD PERSIAN GULF, IRAN

4	08/07/04	Approved For Construction	LG	H.J.Yang	G.C.Lee		S.C.Kim
3	11/06/04	Issued to Company	LG	H.J.Yang	G.C.Lee		S.C.Kim
2	19/09/03	Issued for Company Review	FWEL	S. Jacobs	S. Sim		
1	24/07/03	Issued for Company Review	FWEL	S. Jacobs	S. Sim		S. Sim
0	06/12/00	Issued for Basic Design Package	TECHNIP	V. Zarna	F. Durot		F. Durot
REV.	**DATE**	**DESCRIPTION**	**ORIG.**	**PRPD**	**CHKD**	**APP'D**	**CONTRACTOR's APP'D**

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GENERAL STATEMENT

The attached TOTAL EXPLORATION PRODUCTION (TEP) General Specification n° SP - TCS - 922 Rev. 1 is confirmed as the Project Specification, except as amended herein, and renumbered/reissued as specification n° RP 6340 S 999 0400 07 Rev. *4* for SOUTH PARS PHASES 9 & 10 Project. △4

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ADDENDUM

The following items refer to the attached TEP General Specification SP -TCS- 922 Rev. 1.
The clauses set out below shall modify or replace the existing clauses in the original specification as noted.

GENERAL COMMENT

• The following terms which are used in the this general specification are deleted and replaced :
Replace :

– Engineering company by contractor : Shall mean the party which has entered under a Contract Agreement with POGC for the design, engineering, procurement and construction of the Onshore facilities regarding the South Pars Development Project, Phases 9 & 10.

– TOTAL or TEP : Shall be read as POGC or his nominated Representative.

– Manufacturer : Shall mean any person, firm or company which manufacture or supply equipment or services for the performance of any item of the work.

– Work : shall mean works and/or services entrusted by the contractor under the Contract with POGC.

– Inspection agency : this term designates the Inspection Bodies which work under the supervision of POGC. In some cases, POGC may also act as the inspection agency.

1. DESIGN CODES

• Delete and replace :

– BS 5500 by : PD 5500.

– OCMA by : EEMUA Publication N° 143.

4. QUALIFICATION TESTS

• Add a fifth line :

– However, the acceptance of any work and/or release of the vessels by

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POGC/contractor and/or the inspection agency shall in no way relieve the manufacturer of any responsibility for carrying out all the provisions of the CODE and this specification.

5. FABRICATION

- Add the following:
  - Witness of any pneumatic test, e.g. nozzle reinforcing pad, if required.
  - Witness of pressure vessel hydrostatic test and approval of relevant charts.

Delete section 7. DOCUMENTATION and replace by:

7. RELEASE FOR SHIPMENT

No vessel shall be released for shipment from MANUFACTURER's shop until it has been approved by INSPECTION AGENCY.

Add the following:

8. DOCUMENTATION

The INSPECTION AGENCY shall use the TOTAL specification no. SP-TCS-921 "Check list for inspection of pressure vessels" as a guide to perform the actual inspection in the MANUFACTURER's shops and as a record of the current status of the inspection of each vessel.

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TOTAL EXPLORATION PRODUCTION

GENERAL SPECIFICATION

PIPING - VESSELS - WELDING

SP - TCS - 922

INSPECTION PROCEDURE

FOR PRESSURE VESSELS

1	February 95	New issue
0	October 87	First issue
REV. N°	DATE	PURPOSE OF THE REVISION

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1 - DESIGN CODES

Design codes covered in this specification are either the BS 5500, the ASME VIII div. 1, or the ASME VIII div. 2, and (for Heat Exchangers only) the TEMA and the OCMA Codes.

2 - DRAWINGS AND CALCULATION NOTES

These documents shall be handed over by the ENGINEERING COMPANY to the INSPECTION AGENCY with an "Approved for Construction" mark-up on them.

Any further revision of these documents must be despatched immediately to the INSPECTION AGENCY by the ENGINEERING COMPANY in the same manner.

3 - MATERIALS

Review of sub-orders and expediting of sub-suppliers are under the responsibility of the ENGINEERING COMPANY. When required in the design code or in the project specifications, the INSPECTION AGENCY shall carry out the shop inspection of all main materials, namely (as applicable) : plates for shells, heads, flanges, tubes, tubes-sheets, nozzles, bolting, etc.

The scope of this inspection is, within the requirements of the design code, to perform part or all of the following activities on those materials, as applicable :

Visual inspection
Dimensional check
Witnessing of samples cut-out for mechanical testing
Witnessing of mechanical tests
Non destructive inspection
Approval of material certificates

(Whether a shop inspection at sub-suppliers is required or not according to the project documents, the INSPECTION AGENCY shall at least review and approve the materials certificates submitted by the MANUFACTURER).

4 - QUALIFICATION TESTS

Approval of Welding Procedure Specifications (WPS) and of their supporting Qualification Test Records (PQR) is under the responsibility of the ENGINEERING COMPANY.

In particular, the acceptance of PQR's already qualified for similar previous jobs is under the responsibility of the ENGINEERING COMPANY and is not acceptable for materials other than ordinary steels at temperature above minus 29 degrees Celsius or ordinary austenitic stainless steels of the 300 series of grades (i.e. 304/316/321, etc.).

The INSPECTION AGENCY shall perform the following :

- Witnessing of welding procedures qualification tests
- Witnessing of welders and operators qualification tests (where necessary)
- Witnessing of fabrication tests (where applicable)

This part of the INSPECTION AGENCY's scope of work not only includes the witnessing of welding operations and of mechanical testing of test coupons, but also the approval of base materials and filler materials as well as the verifications and identifications of materials intended to be used for testing purposes.

Within the scope of the design codes and where such possibility is allowed in the applicable project specifications, the INSPECTION AGENCY may, at its discretion, approve welders and welding operators already qualified for similar and recent jobs.

5 - FABRICATION

The INSPECTION AGENCY shall carry out a sufficient number of visits, and at least on a monthly basis, to carry out the following during the fabrication of the vessels :

- dimensional and visual inspection

- approval of non destructive examination (all NDE is to be performed by the manufacturer but requires the witnessing by or, at least, the approval of the INSPECTION AGENCY)

- verification of welds preparation (fit-up, alignment, bevels, cleanliness, preheating, drying of low hydrogen welding products, etc.)

- verification and identification of filler materials, based on checking against materials certificates produced by the filler materials suppliers

- verification and identification of base materials, based on checking of materials markings against materials certificates already approved by the INSPECTION AGENCY

- review and approval of heat treatment charts

- verification of assembly

- witness of any pressure and/or tightness tests.

6 - FINAL ACCEPTANCE INSPECTION

The INSPECTION AGENCY shall carry out the following final inspection tasks before acceptance of the vessel :

- visual examination
- dimensional check
- stamping
- painting inspection (thickness, quality tests).

7 - DOCUMENTATION

The INSPECTION AGENCY shall use the TOTAL specification n° SP-TCS-921 "Check list for inspection of pressure vessels" as a guide to perform the actual inspection in the MANUFACTURER's shops and as a record of the current status of the inspection of each vessel.